

DIFFERENTIAL EQUATIONS (DEs)

DEs describe how 2 variables are changing

DEs have derivatives $\frac{d''''}{d''''}$

e.g. $\frac{dy}{dx} = 2x^2$ ← DE

Solve by integrating

$$y = \frac{2x^3}{3} + C$$

Solution to a DE is an equation

Solutions with a constant e.g. $+ C$ are called GENERAL Solutions

If extra info is given we can find the PARTICULAR solution

THE METHOD OF SEPARATING VARIABLES

e.g. solve $\frac{dy}{dx} = xy$

Extra info.

when $x = 2$, $y = 100$

Rearrange so that all

y's are on one side and all x's on the other

Both dy and dx must be numerators

$$\frac{dy}{dx} = xy$$

$$\frac{dy}{y} = \frac{xy \, dx}{y}$$

$$\frac{1}{y} dy = x \, dx$$

Now integrate each side

$$\int \frac{1}{y} dy = \int x dx$$

$$\ln|y| = \frac{x^2}{2} + c$$

constant

$$y = e^{\left(\frac{x^2}{2} + c\right)}$$

$$y = e^c \times e^{\frac{x^2}{2}}$$

$$y = A e^{\frac{x^2}{2}} \quad (\text{where } A = e^c)$$

This is the general solution

Find A

when $x = 2$, $y = 100$.

$$100 = A e^{\frac{2^2}{2}}$$

$$100 = A e^2$$

$$A = \frac{100}{e^2} = 13.53352 \dots$$

$$A = 13.5$$

So, $y = 13.5 e^{\frac{x^2}{2}}$

This is a particular solution

• Find the general solution to $\frac{dy}{dx} = y^2$

$$\frac{dy}{y^2} = dx$$

$$\int \frac{1}{y^2} dy = \int 1 dx$$

$$\int y^{-2} dy = \int 1 dx$$

$$\frac{y^{-1}}{-1} = x + c$$

$$\frac{1}{y} = -x + d$$

$$y = \frac{1}{d-x}$$

$$\frac{dy}{dx} = x\sqrt{y}$$

$$dy = x\sqrt{y} dx$$

$$\frac{dy}{\sqrt{y}} = x dx$$

$$\int \frac{1}{y^{\frac{1}{2}}} dy = \int x dx$$

$$\int y^{-\frac{1}{2}} dy = \int x dx$$

$$\frac{y^{\frac{1}{2}}}{\frac{1}{2}} = \frac{x^2}{2} + C$$

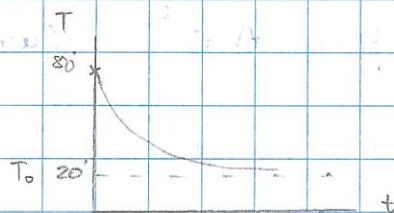
$$2\sqrt{y} = \frac{x^2}{2} + C$$

$$\sqrt{y} = \frac{x^2}{4} + \frac{C}{2}$$

$$y = \left(\frac{x^2}{4} + \frac{C}{2} \right)^2$$

$$y = \left(\frac{x^2}{4} + d \right)^2$$

Example Newton's law of cooling



"rate of cooling is proportional to the difference between current temp. and room temp." (the rate is going down)

$$\frac{dT}{dt} \propto -(T - T_0)$$

$\uparrow$ constant
 $\downarrow$ proportional

$$\hat{y} \propto \hat{z} \quad y = kz$$

$$\hat{y} \propto \frac{1}{\hat{x}} \quad y = \frac{k}{x}$$

$$\frac{dT}{dt} = -k(T - T_0)$$

$\downarrow$ constant of proportionality

E.g. Horse shoe removed from a forge initially 420°C left in a room with constant temp of 15°C . After 20 mins it has cooled to 30°C . How long until it reaches 60°C

$$\text{Given } \frac{dT}{dt} = -k(T - T_0)$$

Ans:

$$\frac{dT}{dt} = -k(T - T_0)$$

t	T
0	420
20	30

separate

$$dT = -k(T - 15) dt$$

integrate

$$\int \frac{1}{T-15} dt = \int -k dt$$

$$\ln(T-15) = -kt + C$$

$$T-15 = e^{-kt+C}$$

$$\rightarrow (e^{-kt})(e^C) \quad \text{let } A = e^C$$

$$T-15 = A e^{-kt}$$

$$T = A e^{-kt} + 15 \quad \rightarrow \text{general solution, Find A, k}$$

use $t=0 \quad T=420$

$$420 = A e^0 + 15$$

$$A = 405$$

use $t=20 \quad T=30$

$$30 = 405 e^{-20k} + 15$$

$$e^{-20k} = \frac{15}{405}$$

$$-20k = \ln\left(\frac{15}{405}\right)$$

$$k = 0.16479 \approx 0.165 \quad \text{3sf}$$

$$\text{So, } T = 405 e^{-0.16479 t} + 15$$

Particular solution

when $T=60$ solve

$$60 = 405 e^{-0.16479 t} + 15$$

$$e^{-0.16479 t} = \frac{45}{405}$$

$$t = \ln\left(\frac{45}{405}\right) \div -0.16479 = 13.3334...$$

$$t = 13.3 \text{ minutes}$$